

Firm Expansion and Stock Price Momentum^{*}

forthcoming in the Review of Finance

Peter Nyberg[†]

Salla Pöyry[§]

June 24, 2013

Abstract

We document a significant and robust connection between firm-level asset changes and return momentum. Momentum profits are large and significant for firms that have experienced large asset expansions or contractions, whereas they otherwise are small and often insignificant. The interaction pattern is not subsumed by previously documented drivers of momentum and shows up in market states where prior literature has documented an absence of momentum profits. Furthermore, we find a positive time-series relationship between aggregate asset growth and return momentum, and the effect of aggregate asset growth is stronger than that of variables related to business cycles and investor sentiment. While most existing models of firm investment and momentum cannot explain our results, recent real options models appear to hold the most promise.

^{*}JEL Classifications: G12. Keywords: Asset pricing, balance sheet asset growth, momentum

[†]We received valuable comments from seminar participants at the Finnish Graduate School of Finance winter workshop, at the Nordic Finance Network research seminar in Lund, at the FMA Europe conference in Hamburg, and at the FMA conference in New York. We also thank the editor (Burton Hollifield), an anonymous referee, Joon Chae, Dirk Hackbarth, Björn Hansson, Timothy Johnson, Petri Kyröläinen, Petya Platikonova, Paolo Sodini, Raman Uppal and Ning Zhu for their comments. An Internet Appendix detailing further results is available from the corresponding author's website.

[‡]Corresponding author. Aalto University School of Business, Department of Finance, P.O. Box 1210, 00101 Helsinki, Finland. Email: peter.nyberg@aalto.fi.

[§]Hanken School of Economics. Dept. of Finance and Statistics, P.O. Box 479, 00101 Helsinki, Finland.

1 Introduction

Few stock market anomalies have received such vast attention among researchers as the momentum effect first documented by Jegadeesh and Titman (1993). Still, after almost two decades of its initial discovery, the momentum anomaly remains an intellectual curiosity: a simple trading strategy that buys stocks with the highest returns over the past three to 12 months and sells stocks with the lowest returns over the same horizon produces profits that remain large after standard adjustments of risk (see e.g. Fama and French, 1996). The prevalence and robustness of the momentum effect justify the abundance of theoretical and empirical research that has been directed at uncovering the underlying reasons for the large payoffs from the trading strategy.¹

In this paper, we investigate the role of firm-level asset expansion as a source of momentum profits. Our paper is motivated by the growing literature that models the relation between firm-level investment decisions and expected returns.² Although existing studies do not formally model the ability of asset expansion to predict momentum profits, they make indirect predictions about the impact of changes in firms' asset bases on expected return momentum. Given the interactions implied by prior literature, the question that emerges is how the different predictions are supported by data. To the best of our knowledge, our paper is the first to explicitly study the interaction between firm-level asset changes and momentum effects.

Our analysis shows that the momentum effect is closely connected to the investment behavior of firms. We show that large changes in the asset base of a firm enhances short-term stock return momentum even when controlling for previously documented firm-level drivers of momentum. Our results reveal a momentum pattern where profits from the strategy are essentially

¹Barberis, Shleifer, and Vishny (1998), Daniel, Hirshleifer, and Subrahmanyam (1998) and Hong and Stein (1999), among others, present behavioral explanations for momentum. On the other side of the spectrum, Berk, Green, and Naik (1999), Johnson (2002) and Sagi and Seasholes (2007) introduce theoretical models that predict price momentum without appealing to behavioral arguments.

²Relevant papers within this strand of literature include Berk et al. (1999), Gomes, Kogan, and Zhang (2003), Zhang (2005), Chen, Novy-Marx, and Zhang (2010), as well as Hackbarth and Johnson (2012).

confined to firms that have experienced a large contraction or expansion in their total assets. The strong empirical evidence in our paper has significant implications for models aiming to uncover the sources of momentum profits or model the relation between firm-level investments and expected returns.

Existing theoretical investment-based asset pricing models imply various underlying structures in momentum profits that incorporate asset growth. Our results are most closely related to the investment-based asset pricing models by Berk et al. (1999) and Hackbarth and Johnson (2012). In the model of Berk et al. (1999), momentum profits arise due to slow changes in firms' asset bases resulting in staleness in expected returns that lead to momentum. Our main empirical result—that momentum is driven by rapid rather than slow changes in firms' assets—does not seem compatible with this theoretical prediction. On the other hand, our interaction pattern is notably supported by the recent model of Hackbarth and Johnson (2012). They propose a different underlying dynamic that considers the impact of cross-firm heterogeneity in investment flexibility on expected returns. They show that a distinguishing feature of firms with more investment flexibility is that their risk falls, on average, as profitability declines and as operating leverage increases. However, for firms with less investment flexibility, risk rises as profitability declines and as operating leverage increases. Their model implies that return autocorrelations should be U-shaped conditional on lagged operating variables which translates into enhanced momentum returns for firms with large expected changes to their asset base—that is, firms near the exercise of expansion or contraction options. The extremes comprise a significant share of firms that subsequently experience extreme changes in their asset bases. Fama and French (2006) provide evidence from univariate regressions that lagged growth shows strong power to forecast asset growth up to three years ahead. This implies that realized asset growth can be seen as a good proxy for expected growth—thus, linking our results to the model of Hackbarth

and Johnson (2012). While their model does not explicitly take on the interaction between momentum and realized asset growth, it nevertheless implies an autocorrelation pattern that closely conforms with our findings.

Other models show a less direct link between firm-level asset growth and return momentum. Chen et al. (2010) propose a new three-factor model, where one of the factors is an investment-based factor that is closely connected to the asset growth variable that we use. We show that while their model decreases the overall magnitude of momentum profits, it nonetheless fails to explain the differences in profits across asset growth groups. Johnson (2002) presents a theoretical model of momentum, where a firm's log price to dividend ratio is convex with respect to expected growth rates, implying that stock returns are more sensitive to changes in expected growth when expected growth is high. We show, however, that the interaction between asset growth and momentum is not driven by within-group differences in expected growth rates, measured as in Liu and Zhang (2008) by the momentum portfolios' factor loadings on the growth rate in industrial production.

Our results are also connected to those of Cooper, Gulen, and Schill (2008) who examine the cross-sectional relation between firm-level asset growth and future returns. They show that firms with high lagged asset growth rates display anomalously low returns relative to standard adjustments for risk. However, they do not investigate the connection between asset growth and return momentum. We extend their results by showing that in addition to predicting the magnitude of future returns, asset growth is also a strong and independent predictor of future return autocorrelation. The combined evidence on the strong forecasting ability of asset growth on returns provides valuable insights into the complex linkages between risk and investment.

Summing up, existing theoretical and empirical models provide contradictory and incomplete predictions regarding the interaction of asset growth and momentum. By documenting

a strong and robust interaction between asset growth and return momentum, our results have implications for models studying the impact of investments and disinvestments on risk. The value and investment premiums have previously been linked to investment based-mechanisms. Linking the momentum premium and investment can thus potentially give rise to a unified framework simultaneously explaining the economic drivers behind the different risk premiums and thus provide a more comprehensive explanation of stock returns.

In the first step of our analysis, we divide the stocks into ten groups based on their past asset growth rates using the intersection of CRSP and Compustat firms during a sample period of 40 years. We follow the momentum strategy of Fama and French (2008) and assign the stocks within the asset growth groups into five portfolios based on their 11-month past returns. Then, we study the returns of a monthly strategy that buys (sells) the prior 11-month winners (losers) within each of the groups. These independent double-sorts reveal a strong interaction between firm asset expansion and momentum: momentum payoffs are large for firms that have either contracted or expanded heavily, whereas they are small for firms that have not had large changes in their asset bases. In particular, for firms that, on average, show a lagged asset growth rate close to zero, the equal-weighted 5-1 momentum payoff is 0.26% per month and statistically insignificant. Moving up along the asset growth deciles, the momentum profits show an almost monotonically increasing pattern. For firms with the largest past asset growth rates, the momentum profit is 1.52% per month and highly statistically significant. Furthermore, within decile 1, where firms, on average, have contracted by 10%, the monthly average momentum payoff is 1.03% and statistically significant.

We control for a set of previously identified drivers of momentum.³ The positive relationship

³Hong, Lim, and Stein (2000) show that momentum profits are higher for small stocks; Daniel and Titman (1999), Sagi and Seasholes (2007) show that momentum effects are largest for stocks with low book-to-market ratios; Lee and Swaminathan (2000) find that momentum is concentrated in high turnover stocks; Zhang (2006) find that momentum is strongest in stocks with high informational uncertainty proxied by variables such as return volatility; and Avramov, Chordia, Jostova, and Philipov (2007) find that momentum is confined to stocks with

between asset growth and momentum remains strong when we control for market value of equity, book-to-market, share turnover, return volatility and credit rating. We also take into account the criticism by Bandarchuk and Hilscher (2013), who document that several momentum strategies enhance profits simply by trading in stocks with more extreme returns. However, we find that asset expansion is a significant predictor of momentum returns even when controlling for the level of past returns. These results imply that our cross-sectional determinant is highly robust in comparison to many of the previously documented drivers of momentum profits.

We also study the time-series pattern of momentum profits by creating a quarterly measure of aggregate asset growth across firms. Chordia and Shivakumar (2002) and Cooper, Gutierrez, and Hameed (2004) have documented that the momentum effect is dampened during recessions and after periods of negative market returns. These findings are exactly what we would expect based on our cross-sectional results since average firm expansion is smaller during these periods. Thus, if asset growth interacts with momentum, we expect the average firm to show lower momentum profits during recessions and after periods of negative market returns. On the other hand, firms that show large asset expansions in these market states should still show economically significant momentum profits. Our empirical results provide support for this claim. To deepen our understanding of the time series interaction, we assign our sample quarters into four groups based on the magnitude of the aggregate asset growth rates in the previous quarter, and study the quarterly momentum returns during these market states. The results display a dramatic pattern: in periods characterized by low aggregate asset growth rates, the momentum profits correspond approximately to 0.18% per month. When aggregate asset growth increases, the average momentum returns increase monotonically. In periods of high aggregate asset growth rates, the average momentum profits reach their highest level at 1.74% per month.

low credit quality.

As a final test, we regress quarterly momentum profits on lagged aggregate asset growth rates while simultaneously controlling for other time series determinants of momentum profits. The results indicate that asset growth emerges as one of the strongest time series determinants of momentum profits.

Finally, we contrast our findings to the behavioral models of momentum presented in Barberis et al. (1998) and Daniel et al. (1998). We argue that these models do not seem fully compatible with our results. We also investigate the post-earnings-announcement drifts (PEADs) within our asset growth groups. Hirshleifer (2001) emphasizes that if the behavioral biases postulated by Barberis et al. (1998) and Daniel et al. (1998) are at play, we should observe particularly strong PEADs in the groups of stocks that show strong momentum. Our results do not provide much evidence that the PEADs are concentrated among firms that have experienced large asset reductions or expansions. This implies that the large momentum returns in extreme asset growth groups are not driven by the same psychological biases that have been used to explain drifts after earnings announcements.

Taken together, our results provide strong evidence that balance sheet asset growth is one of the key drivers of stock price momentum. Although our results show that momentum returns are closely linked to firm-level investment, we nonetheless acknowledge that return momentum may also arise in part from other unrelated drivers of returns. Asness, Moskowitz, and Pedersen (2012) examine macroeconomic and liquidity risk indicators in an attempt to explain the momentum returns found across various asset classes. However, for equities, it seems natural to consider the momentum implications of systematic risk as a function of firm characteristics. While developing a new theoretical model is beyond the scope of this paper, we conclude that any theory that attempts to explain stock price momentum should also be consistent with the empirical facts linking investment and future returns.

The outline of the paper is as follows. The next section describes our data and presents results from the cross-sectional tests and from the time series tests. We begin by examining the effect of asset growth on momentum and then test the robustness of the interaction when controlling for the previously documented firm-specific and time-series drivers of the return pattern. The third section relates our results to existing risk-based and behavioral explanations of momentum. The fourth section concludes.

2 Interaction Between Asset Growth and Momentum

This section presents our main empirical findings. We begin by describing the data set used in our main tests. Then, we document the cross-sectional relationship between asset growth and momentum. Thereafter, we turn our attention to the time-series relationship of asset growth and momentum profits.

2.1 DATA

We use NYSE, Amex and Nasdaq firms that have data available on the CRSP stock return database and Compustat annual industrial files from 1964 to 2006. To minimize potential back-filling and survival biases, we follow earlier research and require that a firm must be listed on the Compustat files two years before we begin calculating the firm-specific variables. Furthermore, we include only ordinary common equity (CRSP share codes 10 or 11) in our sample. To ensure that the accounting variables are in the investors' information set when the portfolios are formed, we follow the timing convention used in earlier studies. That is, the accounting variables for fiscal year ending in calendar year t are matched with portfolios that are formed at the end of June in year $t + 1$.

Following Cooper et al. (2008), we use a comprehensive measure of investment capturing

all components of growth in the balance sheet as our test variable. Our asset growth variable is defined as the yearly percentage change in total balance sheet assets (Compustat data item 6). For a given firm i , the asset growth variable that is matched with returns from July of year $t + 1$ to June of year $t + 2$ is given by

$$AG_{i,t} = \frac{AT_{i,t} - AT_{i,t-1}}{AT_{i,t-1}}, \quad (1)$$

where AG_i is the asset growth for firm i , and AT_i is the value of the firm's total assets.

As in Fama and French (2008), our portfolio sorts pertaining to stock price momentum are based on a $J/K/L$ strategy, where $J = 11$, $K = 1$ and $L = 1$. That is, we sort stocks into groups based on the magnitude of their returns during a formation period of 11 months. To mitigate the effects of short-horizon negative return autocorrelation (Jegadeesh, 1990) and bid-ask bounces, we skip one month between the formation and holding periods. The resulting momentum portfolios are held for one month; thereafter, the sorting procedure is repeated. We also document the robustness of our results by extending the holding period to $L = 3$ and $L = 6$.

To allow for a meaningful comparison of the momentum effect within the different asset growth groups, we use the same prior return breakpoints within each asset growth group in sorting stocks into momentum portfolios. We use the NYSE prior return breakpoints to assign a stock to a portfolio.⁴

To ensure that tiny and illiquid stocks do not drive our results, we exclude from our main tests stocks whose market value at the end of each June falls below the 20th percentile of market cap of NYSE stocks. This excludes from our main sample the stocks that Fama and French (2008) define as microcaps. In our setup, including the microcaps would distort the equal-

⁴Using prior return breakpoints constructed from all stocks in our sample does not change our conclusions. The prior (2-12) NYSE return breakpoints, the NYSE market equity breakpoints and data on the Fama-French factors are obtained from Kenneth French's homepage.

weighted momentum returns because prior losers within the lowest asset-growth group consist predominately of stocks with a very low market capitalization. These firms then dominate the equal-weighted momentum returns in the lowest asset growth group. However, because the momentum effect is negative for the very smallest firms (Hong et al. 2000), this leads to a negative equal-weighted momentum profit within the lowest asset growth decile. We are therefore motivated to have a screen for market value to mitigate the effect that the very smallest firms have on our results. Nevertheless, we also document that our results hold qualitatively within the groups of microcaps, small stocks (having market values between the 20th and 50th NYSE percentiles) and large stocks (having market values above the 50th NYSE percentile).

2.2 UNIVARIATE SORTS ON ASSET GROWTH AND PRIOR RETURNS

We begin our analysis by documenting descriptive statistics for asset growth deciles and for five momentum portfolios in isolation. First, we investigate 10 portfolios based on univariate sorts on asset growth. Then, we move our analysis to 5 portfolios based on univariate sorts on prior returns.

At the end of each June, using the screen for market value detailed above, we sort our sample of stocks into ten portfolios based on their asset growth rates during the previous year. These portfolios are held for the subsequent 12 months, and then they are rebalanced. The sample period for the holding period returns is from July 1968 to June 2006.

[INSERT TABLE I]

Panel A of Table I shows results for the asset growth deciles. Consistent with Cooper et al. (2008), the average monthly holding period returns decrease as we move from the low to the high asset growth groups. However, in contrast to Cooper et al. (2008), the negative

relationship between asset growth rank and holding period returns is not monotonic.⁵ For the equal-weighted (value-weighted) portfolios, the mean raw return difference between the extreme low and high asset growth portfolios is 0.66% (0.53%) per month, with a Newey-West (1987) t-statistic of 4.24 (3.02). Adjusting the 1-10 hedge portfolio returns with the Fama-French (1993) three-factor model drives down the alpha on the equal-weighted 1-10 spread to 0.45% per month. The corresponding spread between value-weighted portfolios is 0.30%. Both return differences are significant at the 5% level. The table also shows the time-series averages of value-weighted asset growth rates and book-to-market ratios for the portfolios. By construction, the average asset growth rates increase as we move along the asset growth deciles. The last row displays descriptive statistics on the time-series averages of value-weighted past returns of the stocks included in the portfolios.

Table I, Panel B documents results for 5 portfolios sorted on the past 11-month returns using NYSE prior (2-12)-return breakpoints, with a one-month lag between the formation and holding periods. The holding period is one month, resulting in an 11-1-1 strategy. The numbers confirm the momentum anomaly in our sample. The holding period EW return spread between past winners and losers is 0.90% per month and statistically significant. Controlling for the Fama-French factors exaggerates the momentum anomaly; the spread in FF3 alphas is 1.15%, with a t-value close to 5. The same conclusion applies to VW returns, even though the momentum effect is slightly reduced.

⁵Similar to our results, Chan, Karceski, Lakonishok, and Sougiannis (2008) document that when small stocks are excluded from the analysis, the asset growth effect shows up only in the decile of stocks with the highest asset growth rates. Using all stocks without any screens for size, we are able to replicate the monotonic negative relationship between asset growth rates and portfolio returns, as in Cooper et al. (2008).

2.3 THE CROSS-SECTIONAL INTERACTION BETWEEN ASSET GROWTH AND MOMENTUM

We now turn our attention to the profitability of momentum strategies *within* the asset growth deciles. At the end of June each year, stocks are sorted into ten groups based on their asset growth rates in the previous year. Then within each asset growth decile, we rank stocks at the end of each sample month based on their past 11-month returns (excluding the month preceding the first holding period month) and then group these stocks into 5 portfolios based on the NYSE prior return breakpoints. This results in a 10×5 independent sort on asset growth rates and past returns.

[INSERT TABLE II]

We first concentrate on the 11-1-1 momentum strategy. Panels A.I and A.II of Table II display average EW returns and average VW returns, respectively, on the five portfolios sorted on past returns. We also report the average return on the 5-1 hedge portfolio that takes a long position in past winners and a short position in past losers for each asset growth group.

The momentum profits within the asset growth groups reveal two distinctive patterns. First, momentum seems to be concentrated among firms that have either experienced asset contraction (asset growth decile 1, where firms, on average, have experienced a negative asset growth rate of 10%, see Table I, Panel A) or large asset expansion (asset growth deciles 7 to 10). In particular, firms in deciles 2 and 3, where the asset growth has been close to zero, do not show a reliable momentum effect. Second, moving from asset growth decile 2 to 10, the average momentum profits for equal-weighted portfolios increase almost monotonically, from 0.26% to 1.52% (0.02% to 1.34% when returns are value-weighted). Furthermore, five out of the ten VW and three out of the ten EW raw momentum profits are insignificant, and these insignificant profits are

concentrated in the low asset growth groups. The FF3-adjusted momentum profits are generally significant.⁶

In the results above, we have omitted stocks from our analysis that fall below the 20th NYSE market cap percentile. Panel B.I of Table II replicates the results in Panel A without any screen for size. To preserve space, we report only the 5-1 raw momentum spreads and their t-values.

A marked difference between the EW and VW portfolio returns emerges when we include microcaps in the analysis. The EW momentum spread for decile 1 is negative, albeit insignificant, whereas the corresponding VW spread is positive and highly significant. As previously mentioned, the reason for this discrepancy is that the past loser portfolio in the lowest asset growth deciles contains a disproportionately large amount of microcaps. Hong et al. (2000) document that, within the lowest NYSE/AMEX size decile, momentum profits are negative. Thus, the EW returns for the lowest AG deciles in our Table II, Panel B.I reflect this pattern in the data. The VW returns that control for this size effect confirm the earlier results from Panel A. That is, the momentum profits are large in magnitude and significant in the negative AG growth deciles and increase across the positive asset growth deciles. Thus, our main results in Panel A do not appear to be driven by the screen for market capitalization that we have used.

The results in Table II, Panel A are based on a holding period of one month. However, as it is also customary to study momentum profits on portfolios that are held for longer periods, we replicate the results in Panel A using holding periods of three and six months. Following Jegadeesh and Titman (1993), we construct portfolios with overlapping holding periods. For example, when a holding period of three months is used, we have three observations on a given

⁶The results conditioned on annual accounting data imply, in line with previous studies, that we match accounting data for fiscal years ending in calendar year t with returns from July of $t + 1$ to June of $t + 2$. In an unreported test of robustness, we shorten the lag between the holding period and the conditioning variable by redoing the sorts using quarterly accounting data from Compustat. We confirm that the results are qualitatively unchanged from the sorts based on annual data.

portfolio for any given month. The portfolio return for that month is then calculated as an equal-weighted average of the portfolio constructed that month and the portfolios constructed in the two previous months.

Table II, Panel B.II shows that our conclusions are not materially affected with a longer holding period.⁷ For holding periods of three and six months, the highest momentum returns show up in the high asset growth decile. However, the large momentum profits for asset growth decile 1 that were evident for the one-month holding period appear to have dissipated quicker than the profits in the other deciles.

Avramov et al. (2007) document that distressed firms with a low credit rating tend to show the highest momentum profits. Recall that asset growth decile 1 consisted of firms that, on average, have experienced asset contractions. Hence, a potential concern is that our lowest asset growth decile mainly consists of distressed firms with a low credit rating, and these firms are driving the momentum profits that we have documented. To mitigate this concern, we collect Standard and Poor's Long-Term Domestic Issuer Credit Ratings from Compustat for firms included in our sample. Due to the non-availability of S&P ratings in the early part of the sample, our sample begins in July 1986. The first column in Table II, Panel C replicates the results in Table II, Panel A.I. for this shorter sample and displays the momentum profits across the asset growth groups using all rated and unrated firms while applying the 20% screen for size. The mean momentum profit in decile 1 between July 1986 and June 2006 is 1.00% ($t=3.04$), and the momentum profits across the asset growth deciles show qualitatively the same pattern as in the full sample. Next, we drop all firms that have a credit rating lower than "B" at the time of portfolio formation, implying that we drop firms that fall below the lowest tercile of credit ratings assigned by S&P. With this filter, the mean momentum profit within decile 1

⁷To conserve space, we only report EW returns throughout the rest of our paper. VW returns produce similar results and are available upon request.

increases slightly to 1.02% ($t=2.92$). The remaining columns show the results when we apply our portfolios sorts to only those firms that have a credit rating ('Only rated') and then dropping firms rated below B ('Only rated, excluding rated below B').⁸ Among the rated firms, the mean momentum profit for asset growth decile 1 is 0.62% ($t=1.82$), and the exclusion of low-rated firms increases the profit slightly to 0.67% ($t=1.79$). Thus, while rated firms show a relatively smaller momentum effect in asset growth decile 1 than firms in the unrestricted sample, the high momentum profit in the low asset growth decile does not appear to be driven by distressed firms having a low credit rating.

2.4 CONTROLLING FOR PREVIOUSLY DOCUMENTED DRIVERS OF MOMENTUM

Prior literature has documented several firm-specific characteristics that appear to be connected to the magnitude of momentum profits. For example, Jegadeesh and Titman (1993) and Hong, Lim, and Stein (2000) show that the returns on small firms exhibit larger momentum effects than the returns on large firms. Daniel and Titman (1999) and Sagi and Seasholes (2007) document that momentum seems to be concentrated within the group of firms having low book-to-market (BM) equity ratios. Lee and Swaminathan (2000) find that firms with high trading volumes show relatively larger momentum, whereas the results in Zhang (2006) and Arena, Haggard, and Yan (2008) indicate that momentum profits are high among firms with high past return volatility. Furthermore, Avramov et al. (2007) report that momentum seems to be concentrated in firms that have high credit risk. In what follows, we show that the momentum profits within the asset growth groups are not driven by the previously documented interaction variables.

⁸On average, 51% of firms in our sample have a credit rating at the time of portfolio formation, implying that we lose about half of the stocks in our portfolios. The lower level of portfolio diversification is also reflected in the relatively smaller t-statistics obtained for this sample.

To purge the effect of the control variables, we start with independent three-way portfolio sorts. First, we sort our sample of stocks into three groups based on the magnitude of the control variable. Then, in an independent sort, we arrange the stocks into three groups based on their asset growth rates. Each stock is then allocated to one of the nine resulting control-variable / asset-growth groups. Finally, within each of the nine groups, we rank stocks at the end of each sample month based on their past 11-month returns (excluding the month preceding the first holding period month) and then group these stocks into 5 portfolios based on the NYSE prior return breakpoints. This allows us to study the returns on the 11-1-1 momentum strategies while simultaneously controlling for a given control variable and asset growth rates. If, for example, the interaction between asset growth and momentum were simply driven by the fact that high asset growth firms are characterized by low BM ratios, we would not expect to find a spread in momentum profits between high and low asset growth firms when BM ratios are held relatively constant.

A detailed description of our portfolio sorts and the included variables can be found in the Internet Appendix. To obtain a sufficient number of stocks in each of the independently sorted 45 portfolios, we use all the stocks in our sample without applying a screen for market value. We report EW returns and corresponding t-statistics from a 5-1 strategy that buys the prior winner portfolio and shorts the prior loser portfolio for each control-variable/asset-growth bin. The results using VW returns, which downsize the possible dominance of microcaps in some portfolios, are qualitatively the same. These results, and some further robustness checks, are presented in the Internet Appendix. The results from the portfolio sorts are reported in Table III, Panel A.

[INSERT TABLE III]

To control for market value, we independently sort the stocks into three size and asset growth groups at the end of each June. Following Fama and French (2008), the size breakpoints are defined as the NYSE 20th and 50th percentiles of market cap for NYSE stocks. Then, within each of the nine size/asset-growth groups, we study the monthly average return on the 11-1-1 momentum strategy. Table III, Panel A.I shows the results for each size/asset-growth bin. The last row of the panel displays the difference in momentum profits between high and low asset growth groups within each size group. For all size groups, the high asset growth group displays the largest momentum profits.⁹ Within the group of large stocks, only the high asset growth group shows a reliable momentum effect.

Next, in Panel A.II of Table III, we control for book-to-market and find that the high asset growth groups once again show the highest momentum profits, and all of the differences in momentum profits between high and low asset growth groups are significant. Similarly, the portfolio sorts with asset turnover in Panel A.III show that the high asset growth stocks show higher momentum profits than low asset growth stocks within each turnover group.¹⁰ In controlling for idiosyncratic volatility, we calculate the measure relative to the Fama-French (1993) three-factor model for each stock in the sample using daily returns within the 11-month portfolio formation period. As observed from Table III, Panel A.IV, the asset growth interaction with momentum is still significant when controlling for lagged return volatility.

We conclude the portfolio sorts by controlling for credit risk in Panel A.V of Table III. Following Avramov et al. (2007), we collect Standard and Poor's Long-Term Domestic Issuer

⁹The difference in momentum profits between high and low asset growth groups is significant within microcaps and large stocks but insignificant within the group of small stocks. The low asset growth group within small stocks contains a subgroup of stocks with a large negative asset growth rate. Due to the U-shaped asset growth-momentum relationship, demonstrated in Panel A of Table II, firms that have experienced a large asset contraction tend to display high price momentum. This drives up momentum profits within the low asset growth groups. Using value-weighted returns, the difference is significant within each size group.

¹⁰As in Lee and Swaminathan (2000), we exclude Nasdaq stocks from our sample due to the double counting of dealer trades and report results only for the NYSE/Amex sample.

Credit Ratings from Compustat and match these credit ratings with the stocks in our sample. Restricted by the availability of credit ratings from Compustat, our portfolio sorts start from July 1986. The results confirm that the asset growth interaction survives even when controlling for credit rating. Within each credit rating group, the differences in momentum profits between high and low asset growth groups are positive and statistically significant.

2.4.1 Fama-MacBeth Regressions

The portfolio sorts indicated that the effect that asset growth has on momentum profits survives the inclusion of the control variables. However, there are some shortcomings with the robustness tests in the previous section. In essence, we would like to know the marginal significance of asset growth for momentum profits with respect to the previously documented interaction variables. Unfortunately, the universe of NYSE/AMEX/Nasdaq stocks with available historical accounting data in Compustat is not sufficiently large to evaluate the joint effect of the control variables on the asset growth-momentum relationship through four- or five-way portfolio sorts.

To circumvent the problem, we run firm-level Fama-MacBeth (1973) regressions of monthly stock returns on empirical determinants of cross-sectional returns and on terms that capture how the variables interact with momentum. The full cross-sectional model that we estimate each sample month is given by

$$\begin{aligned}
 r_{i,t} = & b_0 + b_1 AG_i + b_2 \log(MV_i) + b_3 \log(BM_i) + b_4 ret11_i + b_5 IVOL_i + b_6 TO_i + b_7 CR_i \\
 & + b_8 (ret11_i \times AG_i) + b_9 (ret11_i \times \log(BM_i)) \\
 & + b_{10} (ret11_i \times TO_i) + b_{11} (ret11_i \times CR_i) + e_{i,t}.
 \end{aligned}
 \tag{2}$$

The dating conventions follow our previous definitions. The monthly returns from July year

t to June year $t + 1$ are matched with the asset growth rate (AG) measured during the fiscal year ending in calendar year $t - 1$. The log market value of equity, MV , is measured at the end of June in year t . Log book-to-market equity, BM , is based on accounting data from the fiscal year ending in calendar year $t - 1$, whereas the market value in the denominator is from June of year t . The variable $ret11$ is the lagged 11-month return used in our earlier portfolio sorts, measured from month $t - 2$ to $t - 12$. The average turnover (TO) and idiosyncratic volatility ($IVOL$) are measured over the same period as the lagged 11-month return. CR is the most recent credit rating assigned to the firm.¹¹ The coefficients for the interaction terms, b_8 to b_{11} , measure how the past return interacts with the control variables. If, for example, b_8 is positive and significant, a higher asset growth rate increases the effect that lagged past returns have on holding period returns, i.e., there is a stronger price momentum effect when the change in total balance sheet assets is large.

In the regression results reported below, we apply the 20% screen for size to the stocks in our sample and exclude stocks with a negative BM ratio. As in the portfolio sorts, we exclude Nasdaq stocks from our sample when the turnover variable is included. Furthermore, the sample starts from July 1986 for the estimations that contain the credit rating variable. The standard errors are based on Newey-West (1987) estimates.

Regression models with interaction variables are potentially plagued by multicollinearity. We find the largest correlation between the past returns and the interaction term between log market value and past returns (the average correlation between $ret11$ and $ret11 \times \log(MV)$ is 0.99), the second largest average absolute correlation is between past returns and their interaction with idiosyncratic volatility (0.92). Thus, we exclude the size and volatility interactions from our model above. The correlations between the other variables are considerably lower.

¹¹We linearly transform the 22 credit rating groups to a scale from -1 to 1, where -1 denotes the firms with the highest credit rating ("AAA"), and 1 denotes the firms with the lowest rating ("D").

Table III, Panel B, columns 1 and 2 show the results without interaction variables. The signs of the parameter estimates do not produce any surprises relative to prior literature. Consistent with Cooper et al. (2008), lagged asset growth shows up with a negative and highly significant coefficient. Furthermore, book-to-market and lagged 11-month returns display significantly positive coefficients, whereas the coefficient for market value is negative. These results are perfectly in line with the well-documented value, momentum and size effects. As in Ang, Hodrick, Xing, and Zhang (2006), the sign of *IVOL* implies that higher idiosyncratic volatility is associated with lower returns. In the NYSE/AMEX sample (column 2), high turnover is associated with significantly lower returns.¹²

Next, we investigate the results from regressions that include the interaction terms. Specification 3 adds the interaction between asset growth and past returns to our baseline model presented in column 1. The coefficient for the interaction term is positive and significant ($t=2.86$). Including the interaction between past returns and book-to-market ratio (specification 4) does not drive away the significance of the asset growth interaction. Indeed, the significance of the asset growth interaction remains high ($t=2.22$), whereas the book-to-market interaction is insignificant ($t=-1.43$). Column 5 further demonstrates the robustness of the asset growth interaction: the interaction remains significant even when the specifications control for the interaction of past returns with turnover.

The two remaining specifications control for the interaction between credit rating and past returns. Recall that in these tests, our sample length is cut in half, and our regressions begin from July 1986. Results from specification 6 reveal that the credit rating interaction, while having the right sign relative to portfolio sorts, does not reach statistical significance ($t=1.50$). The asset growth interaction, on the other hand, remains significant, with a t -value of 2.12.

¹²A similar pattern has been documented, among others, by Datar, Naik, and Radcliffe (1998) and Lee and Swaminathan (2000).

Finally, in specification 7, we test for the significance of the asset growth interaction while simultaneously controlling for the previously documented drivers of momentum. From these results, asset growth emerges as the variable with the strongest amplifying effect on return momentum. The asset growth interaction displays a t-value of 2.78, whereas the other interactions are rendered insignificant.

As a whole, our results thus far indicate that firm-level asset growth is connected to the profitability of momentum strategies. More significantly, when we control for a set of previously documented drivers of momentum, asset growth emerges as the strongest and most significant interaction variable that predicts the magnitude of the momentum effect.

2.4.2 Controlling for Dispersion in Past Returns

Bandarchuk and Hilscher (2013) present a critique of previous studies that have relied on double-sorts to document how momentum profits vary with various firm characteristics. Their critique is based on the observation that firms with extreme characteristics have more extreme past returns. Because momentum profits across portfolios increase with extreme past returns, the previously documented interaction patterns merely emerge as a result of differences in the dispersion of past returns across the portfolios. For example, firms with high turnover display more dispersion in past returns than firms with low turnover. Thus, it is not ex ante surprising that momentum strategies within high turnover groups produce larger profits than momentum strategies within low turnover groups. The authors subsequently show that most of the previously documented momentum interactions appear to only capture the effect that extreme past returns have on momentum profits.

We are sympathetic to the concern raised by Bandarchuk and Hilscher (2013). We note, however, that the Fama-MacBeth regressions in the previous subsection suggest that our asset

growth interaction is not merely driven by differences in past returns across the asset growth groups. The estimated interaction coefficients control, to some extent, for the spread in past returns: they measure the magnitude at which the sensitivity of holding period returns to a fixed change in past returns varies across the asset growth spectrum. Nevertheless, to alleviate any remaining concerns, we present in our Internet Appendix results from a further robustness test that directly controls for the dispersion of past returns within asset growth deciles. The results, based on benchmark-adjusted momentum returns (as in Daniel, Grinblatt, Titman, and Wermers, 1997), demonstrate that the interaction between asset growth and momentum is not explained by a larger dispersion in formation period returns in the extreme asset growth deciles.

2.5 TIME-SERIES EVIDENCE

The previous sections indicated that firm-level balance sheet asset growth is a strong cross-sectional determinant of momentum profits. We now turn our attention to the time series of momentum profits.

2.5.1 *Momentum Profits during Different Business-Cycle Periods*

Chordia and Shivakumar (2002) document a business-cycle pattern to momentum profits where momentum profits are positive during expansionary periods but negative (though insignificant) during recessions. Our cross-sectional evidence that price momentum is connected to firm expansion is consistent with this finding. During expansionary periods, firm-level expansion is likely to be larger than during recessions.¹³ If momentum is connected to firm expansion, we would expect the average firm to show higher momentum profits during periods that are associated

¹³We confirm this statement by collecting quarterly asset growth data from Compustat and calculating the cross-sectional average asset growth rates for each quarter during our sample period. During recessions, as defined by the NBER, the time-series mean of the average asset growth rate is 2.4% per quarter, whereas it is 3.9% during expansions.

with higher asset growth rates. Furthermore, even if momentum profits on aggregate are lower in recessions due to weaker firm expansion, we would nevertheless expect firms expanding heavily to show positive momentum profits during recessionary periods as well. We now document that this is indeed the case.

[INSERT TABLE IV]

Panel A of Table IV documents the momentum profits during the different business cycle periods. We categorize the holding period months from July 1968 to June 2006 as expansionary or contractionary months based on the classifications made by the NBER. Altogether, there are 391 months that are classified as expansions and 65 months classified as recessions. To investigate the significance of momentum profits in the two economic states, we regress the time series of our 5-1 momentum portfolio returns on an expansionary dummy variable and a contractionary dummy variable without an intercept. The estimated coefficients for the two dummy variables capture the mean momentum profits in the two states.

The first row, labeled 'all stocks', shows the aggregate 5-1 momentum profits for all stocks in our sample. The mean momentum returns in the two periods follow the same pattern as in Chordia and Shivakumar (2002): momentum profits are larger in expansions (mean=0.96%, $t=4.20$) than in recessions (mean=0.54%, $t=0.85$).

The remaining rows in the table show the mean profits of the momentum portfolios within ten asset growth groups. The numbers demonstrate that, during recessions, asset growth groups 2 to 7 show average momentum profits that are close to zero and even negative, whereas firms that expand heavily in recessions have positive momentum profits. In particular, groups 9 and 10, with the largest asset growth rates, show large average monthly momentum profits of 1.05% and 1.40%, respectively. However, these profits are not significant, with respective t -values of

1.37 and 1.52. The lack of statistical significance is likely to be due to the low number of months that are classified as recessions. Nevertheless, the high point estimates of momentum profits within the extreme asset growth groups provide further support for our claim that firm-level expansion is an important driver of momentum.

2.5.2 Momentum Profits Following Up and Down Markets

Cooper et al. (2004) document that momentum profits are significant only when the lagged one- to three-year stock market returns have been positive. This empirical finding appears to be related to our results. Following up-markets, managers are likely to invest more due to better access to capital.¹⁴ This leads to larger firm expansion and, consistent with our cross-sectional results, to larger momentum effects.

Panel B of Table IV presents the momentum profits for the asset growth groups in up and down markets. We use 12-month cumulative returns on the CRSP value-weighted market index as a proxy for market returns. If the 12-month lagged return on the index has been positive (skipping one month between the formation and holding period), we classify a holding-period month as an UP month. Otherwise, the month is classified as a DOWN month. In total, 346 months are UP months, and 110 are DOWN months.

We confirm that the results in Cooper et al. (2004) hold for our sample. For a momentum strategy involving all stocks (first row), the payoff in UP months is a highly significant 1.19% per month ($t=5.80$). In DOWN months, the unconditional strategy essentially yields zero profits, with a mean of -0.02%. Examining the momentum profits within the asset growth groups, the returns in UP markets reveal that the extreme asset growth groups once again show up with the

¹⁴In an unreported test, we regress quarterly average asset growth rates on lagged 12-month market returns and a constant. The t -statistic for the lagged market return is 3.28, and the r -square from the regression is 10%. The correlation between lagged market returns and the quarterly asset growth rate is 0.32. Furthermore, the average quarterly asset growth rate following 12-month DOWN markets is 2.80%, whereas it is 4.32% following 12-month UP markets.

highest profits. However, in DOWN markets, only asset growth decile 10 displays a relatively high average momentum profit of 0.97%, albeit with an insignificant t-value of 1.29. The rest of the asset growth deciles show momentum payoffs that range from -1.35% (decile 3) to 0.13% (decile 8).

Despite the high 0.97% per month time-series mean of the DOWN market momentum profits in asset growth decile 10, the insignificance of the payoff might raise the question if the asset growth/momentum interaction survives in DOWN markets. We believe, however, that the results in Table IV, Panel B are broadly in line with our earlier findings. First, the momentum profits generally show an increasing pattern from asset growth decile 3 to decile 10. Second, it is a well-documented fact that volatility tends to be high in down-markets. The standard deviation of monthly momentum returns within decile 10 is 10.3% in DOWN months, whereas it is only 4.9% in UP months. The high volatility coupled with a low number of DOWN observations leads to a lack of statistical significance. Third, one single month, January 2001, seems to have been devastating for momentum strategies. During this month, the unconditional 5-1 momentum strategy lost -42.1%. We do not wish take a strong position on whether this observation is an outlier (a random event distorting our statistical inference).¹⁵ For completeness, Panel B of Table IV also shows the results for the DOWN month momentum profits, where January 2001 is excluded from our sample. In this case, the mean momentum profit for asset growth decile 10 is 1.52% per month (compared to 0.97% when January 2001 is included) and statistically significant ($t=2.23$), despite the low number of observations. In unreported results, we furthermore test that when we include all DOWN months, the median momentum profit for decile 10 is 1.61%,

¹⁵Using data on 10-1 momentum portfolio returns from Kenneth French's homepage, we find that January 2001 marks the third-largest loss that has occurred on a simple momentum strategy since 1927. The 10-1 momentum loss in this month was -60.4%. The two other outliers occurred in August 1932 (-83.2%) and September 1939 (-89.7%). This largest loss in 1939 occurred after a 12-month UP market, which indicates that these observations are pure outliers and may not be connected to lagged market states. Furthermore, the behavioral theories that motivate the use of lagged market returns as a conditioning variable for momentum profits do not predict that momentum profits should be *negative* when investor overconfidence decreases.

and 60% of the monthly momentum profits are positive. In UP months, the median is also 1.61%, and 63% of the monthly momentum profits are positive. Thus, correcting for one single extreme return, firms that have experienced the largest asset expansions appear to provide similar momentum profits in both UP and DOWN markets.

2.5.3 Momentum Profits Following Periods of Low and High Sentiment

Building on the behavioral theory proposed by Daniel et al. (1998), Antoniou, Doukas, and Subrahmanyam (2010) argue that market-wide investor sentiment should be connected to aggregate momentum profits. Using different proxies for sentiment, they find support for this hypothesis: momentum profits are high after periods of optimistic sentiment and insignificant after periods of pessimistic sentiment. In what follows, we show that the asset-growth/momentum interaction shows up in both pessimistic and optimistic market states.

We use the monthly sentiment index constructed by Baker and Wurgler (2006, 2007) to classify our sample months in pessimistic and optimistic periods.¹⁶ Following Antoniou et al. (2010), we calculate a moving average of the sentiment values for the three months prior to the holding period. If this average sentiment level falls within the bottom 33% of all months, we classify the holding period month as pessimistic. The rest of the sample months are classified as optimistic months.

The first row in Panel C of Table IV replicates the result in Antoniou et al. (2010): for the unconditional 5-1 momentum strategy, profits are high and significant following optimistic periods (mean=1.13%, $t=4.04$) but insignificant after pessimistic periods (mean=0.45%, $t=1.02$). However, despite this insignificance of unconditional momentum profits, firms with high asset expansion still display reliable momentum profits even after periods of pessimistic sentiment.

¹⁶The sentiment index is available from Jeffrey Wurgler's homepage, <http://pages.stern.nyu.edu/~jwurgler/>.

Asset growth decile 10 shows up with a mean momentum profit of 1.03% ($t=2.24$). On the other hand, firms with medium asset expansion (deciles 2 to 6) display mean momentum profits that are close to zero. Thus, even though pessimistic investor sentiment appears to have a dampening effect on the magnitude of momentum profits, it does not significantly affect the interaction between asset growth and momentum.

2.5.4 Momentum Profits Conditioned on Aggregate Asset Growth

We have thus far provided evidence indicating that the interaction between asset growth and momentum survives in various market states, where prior research has documented an absence of momentum profits. We now provide further evidence on the time-series dimension, in support of our premise that asset growth is an independent driver of momentum. If firm-level asset growth is connected to the magnitude of momentum profits, we would expect the magnitude of unconditional momentum profits to be lower in periods of low aggregate asset growth and higher in periods of high aggregate asset growth. Thus, using aggregate firm expansion as a conditioning variable, we expect to find a similar spread in unconditional momentum profits across these market states, as we found in the momentum portfolios that were conditioned on firm-level asset growth.

We collect quarterly balance sheet total assets using the intersection of all firms included in the CRSP and Compustat quarterly databases. For each firm, we calculate the annual asset growth rates using formula (3) below

$$AG_{i,t}^{quarter} = \frac{AT_{i,t}^{quarter} - AT_{i,t-4}^{quarter}}{AT_{i,t-4}^{quarter}}, \quad (3)$$

where $AT_{i,t}^{quarter}$ denotes total balance sheet assets for firm i during the fiscal quarter ending in

calendar quarter t .¹⁷ Then, for each quarter t , we take an average over all firms for which the asset growth rates are available. This produces a measure of aggregate average asset growth for a given calendar quarter.

$$AG_t^{quarter} = \frac{1}{N} \sum_{i=1}^N AG_{i,t}^{quarter} \quad (4)$$

Our aim is to study unconditional momentum profits conditioned on low and high aggregate asset growth rates. To this end, we use returns on 10 momentum portfolios that include NYSE, AMEX and Nasdaq stocks and are formed on NYSE prior (2-12) return decile breakpoints. These portfolio returns are available from Kenneth French's homepage. We calculate quarterly momentum returns by compounding the monthly returns on the loser and winner portfolios over the quarters and then calculating the difference between the compounded winner and loser portfolio returns. We match the quarterly asset growth rates $AG_t^{quarter}$ with the subsequent 10-1 quarterly momentum returns measured during quarter $t + 1$. Then, we sort our time series of quarterly aggregate asset growth rates into four groups based on their magnitudes and study the resulting momentum profits within each of these four market states. Our sample consists of 152 quarterly observations, and each asset growth group contains 38 observations.

[INSERT TABLE V]

Panel A of Table V presents the mean quarterly momentum returns and their corresponding t-statistics when the returns are conditioned on aggregate asset growth. To allow for an easier comparison with our earlier results, the last column shows the quarterly mean returns transformed to a monthly basis.¹⁸ The results display a dramatic pattern. During periods of low aggregate asset growth, momentum returns are small and insignificant. The quarterly mean

¹⁷Note that we use quarterly asset growth rates calculated on a yearly basis (i.e., growth measured over four quarters) to remove predictable seasonal variation that might arise if we used quarter-to-quarter changes. The quarter-to-quarter asset growth rates show some signs of seasonality; a non-parametric Kruskal-Wallis test for the equality of quarterly medians of the asset growth rates rejects the null at a 5% level of significance.

¹⁸That is, the approximate monthly return is given by $r^{month} = 100 \times ((1 + r^{quarter}/100)^{1/3} - 1)$.

is 0.55% (corresponding approximately to 0.18% per month), with an insignificant t-statistic of 0.27. As aggregate asset growth increases, momentum profits follow the same trend and show a monotonically increasing pattern. During periods characterized by the highest aggregate asset growth rates, the momentum profits reach their highest level; the quarterly mean is 5.30% (corresponding approximately to 1.74% per month) and highly significant ($t=4.28$). The difference between quarterly momentum profits in high and low aggregate asset growth periods is 4.76%, with a corresponding t-statistic of 1.97.

In Table V, Panel B, we perform time-series regressions to further evaluate the relationship between aggregate asset growth and momentum profits. The dependent variable is the quarterly 10-1 momentum payoff, whereas the explanatory variables are the one-quarter lagged aggregate asset growth rate and other variables that have previously been shown to predict momentum payoffs. The first model uses the lagged asset growth rate as a sole explanatory variable. The results confirm the finding in Panel A: higher lagged aggregate asset growth rates are connected with larger momentum profits. The coefficient for $AG_{t-1}^{quarter}$ is 0.30 and statistically significant with a t-statistic of 2.45. Following Chordia and Shivakumar (2002), model 2 adds a dummy variable, capturing recessions, to the specification. We code a quarter as recessionary if at least two out of three months within the lagged quarter are classified as recessionary by the NBER. As expected, the point estimate of the dummy variable is negative (-2.63), implying that momentum payoffs are smaller in recessions. The coefficient is, however, not statistically significant ($t=-0.73$). Furthermore, controlling for recessions does not diminish the significance of lagged asset growth rates.¹⁹

Antoniou et al. (2010) show that lagged sentiment levels predict future momentum returns. Therefore, in model 3, we control for the lagged sentiment level, defined as in Table IV.

¹⁹We also estimate model 2 using a contemporaneous dummy variable that captures whether the holding period quarter is recessionary. The results remain robust.

While sentiment is positively related to momentum profits, the relationship is not statistically significant. On the other hand, the effect of lagged asset growth remains significant.

Cooper et al. (2004) document a nonlinear relationship between lagged market returns and momentum profits: momentum profits increase with lagged market returns but decrease with the square of lagged market returns. In model 4, we control for these effects using the lagged 12-month cumulative returns on the CRSP value-weighted market index as a proxy for market returns. The results show that in this specification, the lagged asset growth rate loses its significance ($b=0.14$, $t=1.12$). However, we note that, at least in our sample, extreme downside returns are driving the significance of the specification used by Cooper et al. (2004). Excluding 3% of the lowest market returns from our sample (4 out of 152 observations) makes both the lagged market return and its square insignificant. Furthermore, as Antoniou et al. (2010) note, during these extremely adverse market conditions, momentum strategies cannot easily be implemented due to increased volatility and reductions in liquidity. To study the interaction between asset growth and momentum in "normal" market states, model 5 uses the same specification as model 4 but excludes all negative market observations from the sample. The number of excluded negative market returns is 38, leaving us with 114 observations that are used to estimate model 5. In this specification, lagged asset growth emerges as a highly significant predictor of momentum profits ($t=2.82$), whereas the lagged market return and its square become insignificant, with respective t -values of -1.08 and 0.21 . Overall, while momentum payoffs are better forecasted by lagged market returns than by lagged asset growth rates when all sample points are considered, this relationship between the two explanatory variables is not robust when we control for a few extreme observations or when we study only UP market states. Specifically, in UP market states, the forecasting power of asset growth strongly dominates that of lagged market returns.

3 Discussion

Having established the interaction between asset growth and momentum, we now discuss how our results relate to existing explanations of momentum.

3.1 RELATION TO INVESTMENT-BASED ASSET PRICING

A growing literature studies the relation between optimal investments and expected returns. Most of the literature concentrates on theoretically modeling the relation between firm-level decisions, investment, and the average return patterns related to book-to-market and market capitalization.²⁰ A common prediction in the literature is that firms that invest more should have lower expected returns relative to firms that invest less. However, most of the theoretical models do not explicitly predict that firms that experience large asset changes, either through investment or disinvestment, should show large momentum profits.²¹

An investment-based framework that explicitly studies momentum effects is Berk et al. (1999). In their model, persistence in a firm's assets and systematic risk leads to persistence in expected returns at longer horizons. Contrarian effects at shorter horizons arise because of shocks to the composition of a firm's assets. If slow turnover in firms' assets were the driving force behind momentum, we would expect to see the strongest momentum effects in firms that have not rapidly changed their asset bases. However, our results point to an opposite conclusion. Firms that have experienced rapid shocks to their assets during the previous year, as measured by asset growth, display the strongest momentum effects.

²⁰See, for example, Gomes et al. (2003), Carlson, Fisher, and Giammarino (2004), Zhang (2005), and Cooper (2006).

²¹Such momentum profits could occur if, for example, optimal investment had the tendency to increase the autocorrelation in stock returns, or if the group of firms that are characterized by the largest investment/disinvestment were to show the largest spreads in expected returns.

3.1.1 The Impact of Real Options on Autocorrelations

Hackbarth and Johnson (2012) propose a theoretical model that implies a return structure more compatible with our results. Their framework builds on the impact of cross-firm heterogeneity in investment flexibility on expected returns. Their model assumes that the relationship between realized profitability shocks and expected returns is dependent of the investment and disinvestment options available to the firm. As a result of the real options, risk exhibits a concave region followed by convex region with respect to profitability where the slope in the middle section may have either sign. The effect of good (bad) news on expected returns is nonetheless always more positive (negative) near the exercise of the firm's real options. The momentum implication of the model is that return autocorrelations should be U-shaped conditional on relevant lagged operating variables. With respect to changes in the asset base of firms, the model predicts enhanced momentum returns for firms near the exercise of expansion or contraction options.²² The intuition is that following positive productivity shocks, the firm's risk rises due to more valuable expansion options. On the other hand, when profitability declines, the firm's risk declines due to more valuable contraction options. The model can easily be applied to explain return momentum since profitability or productivity shocks are arguably associated with realized returns in the cross-section. Our findings provide support for the model if one assumes that firms near the exercise of their options comprise a significant share of firms that have experienced extreme changes in their asset bases. Fama and French (2006) provide evidence from univariate regressions that lagged growth shows strong power to forecast asset growth up to three years ahead. This implies that realized asset growth can be seen as a good proxy for expected growth—thus, linking our results to the model of Hackbarth and Johnson (2012). Unlike Berk et al. (1999),

²² Also using real option type models, Sagi and Seasholes (2007) and Garlappi and Yan (2011) similarly predict enhanced momentum profits in restricted samples of stocks. They predict enhanced returns in firms with more growth options and in distressed firms with disinvestment options. The model derived by Hackbarth and Johnson (2012) considers both types of real options resulting in a prediction that is supported by our empirical findings.

their model thus implies an autocorrelation pattern that supports our findings.

3.1.2 Controlling for Profitability

Chen, Novy-Marx and Zhang (henceforth CNZ, 2010) propose a new factor model derived from investment-based asset pricing. Their model consists of the market factor, an investment factor and a return on assets (ROA) factor, resulting in a model for expected returns given below:

$$E[R_i^e] = \beta_{i,MKT} E[R_{MKT}^e] + \beta_{i,INV} E[R_{INV}] + \beta_{i,ROA} E[R_{ROA}], \quad (5)$$

where $E[R_{INV}]$ is the expected premium on a portfolio long in low investment stocks and short in high investment stocks, and $E[R_{ROA}]$ is the expected premium on a portfolio long in stocks with high returns on assets and short in stocks with low returns on assets. They hypothesize that when holding investment-to-assets constant, firms with a high expected ROA should earn higher returns than firms with a low expected ROA. Furthermore, they argue that sorts based on past returns are likely to generate an implied sort on expected ROA because past winners (losers) are more likely to have experienced a positive (negative) shock to their earnings. Consequently, the different factor loadings that winners and losers have on the ROA factor can potentially explain part of the momentum anomaly. CNZ show that their new factor model produces substantially smaller momentum alphas than the Fama-French (1993) three-factor model or the CAPM.

The results in CNZ are connected to our results in the sense that their model contains an investment-based factor that empirically appears to give a reasonable description of the returns on momentum portfolios. While their theoretical framework does not explicitly predict that firms that invest more should show a stronger return momentum, it could be that the winner and loser portfolios in the more extreme asset growth groups show a wider dispersion in their CNZ-factor loadings. It is also plausible that the spread in firm profitability, measured by ROA,

is larger within the more extreme asset growth groups. Next, we examine whether the CNZ model explains the momentum/asset growth interaction.

[INSERT TABLE VI]

For each of the asset growth groups, we regress the time series of the equal-weighted 5-1 momentum returns on a constant, the market factor, and the two CNZ-factors. Data on the CNZ factors are available from January 1972.²³ Table VI, Panel A reports the raw returns on the momentum portfolios, the CNZ alphas and the corresponding t-statistics. The raw momentum profits show the familiar increasing pattern as we move from asset growth group 2 to 10. Turning to the CNZ alphas, our results confirm that their factor model substantially decreases abnormal profits from momentum strategies. For example, the CNZ adjustment cuts down the raw momentum profit in asset growth group 10 from 1.53% to 0.93% per month. Still, abnormal momentum returns within asset growth groups 8 to 10 are large (ranging from 0.75% to 1.02% per month) and statistically significant.

Next, we perform a test where we control more directly for the effects of firm profitability. At the end of June each year, we rank firms in three groups based on their asset growth rates in the fiscal year ending in the previous calendar year. Since we are interested in studying the robustness of the positive interaction between asset growth and momentum, we exclude asset growth decile 1 from these sorts.²⁴ Then, at the beginning of each month, we independently rank firms in three groups based on their quarterly ROAs. ROA is defined as the firm's most recently reported quarterly income before extraordinary items (Compustat item IBQ) divided by the total assets (ATQ) of the firm in the previous quarter. Based on the group ranks, we form nine asset growth/ROA groups. Finally, within each of the nine groups, we study 5-1

²³We thank Long Chen for providing us with the CNZ factors.

²⁴Since the first asset growth decile displays a strong momentum effect, including firms from this decile in our sorts would drive up the momentum profits in the lowest asset growth tercile, thus distorting our inferences.

momentum profits using 5 portfolios sorted on the past 11-month returns within groups where asset growth and profitability are held relatively constant. The holding period is one month. In line with our previous results, we apply the 20% screen for market value. To ensure that we have sufficiently many firms with quarterly accounting data, we start the sample from July 1980.

The results are displayed in Table VI, Panel B. When ROA is held relatively constant, the raw 5-1 momentum profits are largest for the high asset growth groups. For example, for firms that have a high return on assets, the difference in monthly raw momentum profits between the high and low asset growth groups is 0.84% per month (t-value 2.84). The same strong pattern can also be found in the medium ROA group. The pattern in the low ROA group is, however, weaker and the difference in momentum profits does not reach statistical significance.

For all of the ROA groups, the last row in Table VI, Panel B shows the difference between the winner-minus-loser CNZ alpha for the high asset growth group and the winner-minus-loser CNZ alpha for the low asset growth group. If the CNZ factors were to explain the difference between momentum returns in high and low asset growth groups, then we would expect to see an insignificant alpha. However, this is not the case for the medium and high ROA groups. We conclude that the CNZ model provides, at best, only a partial explanation as to why firms with large asset base changes show a stronger return momentum.

3.2 CONTROLLING FOR EXPECTED GROWTH RISK AND MACRO-ECONOMIC VARIABLES

Finally, to assure that the interaction between asset growth and momentum is not a manifestation of some of the previously suggested explanations for the momentum and asset growth anomalies, we test for expected growth risk and macroeconomic risk exposure as suggested by

Johnson (2002) and Cooper and Priestley (2011), respectively. The unreported results from these robustness tests and more detailed descriptions of the estimations are available in the Internet Appendix.

In the first test, we consider the theoretical model by Johnson (2002) where a firm's log price to dividend ratio is convex with respect to expected growth rates. This convexity implies that the prices of stocks with high expected growth are more sensitive to changes in expected growth than prices of firms with low expected growth, and as a consequence, growth rate risk rises with expected growth rates. If exposure to growth rate risk is positively priced, then expected returns should also rise with growth rates. Firms with poor past performance are likely to have had negative growth shocks, whereas firms with good past performance are likely to have had positive growth shocks. Consequently, due to their higher expected growth rates, past winners are riskier and should have higher expected returns than past losers, leading to momentum. Insofar as firms that have experienced high asset growth in the past are affected more than firms with lower asset growth by the growth rate risk as defined in Johnson (2002), the convexity of the log price to dividend ratio in Johnson's model could provide a rationale for the momentum interaction that we document.

As the expected growth risk appearing in Johnson's model is unobservable, we follow the methodology in Liu and Zhang (2008) to operationalize our tests. We estimate the loadings on growth rate in industrial production (MP) for our 50 momentum portfolios controlling for the other Chen, Roll and Ross (1986; henceforth CRR) factors.²⁵ If, as argued by Liu and Zhang (2008), the MP factor summarizes aggregate changes in expected growth, and if growth-related risk were more important within the high asset growth groups thus driving their higher momentum profits, then we would expect to see a larger spread in MP loadings between past

²⁵We thank Laura Xiaolei Liu and Lu Zhang for making the data on the CRR-factors publicly available. The factors include unexpected inflation (UI), change in expected inflation (DEI), the term premium (UTS) and the default premium (UPR).

winners and losers in the high asset growth groups than in the low asset growth groups. However, the spreads in factor loadings between past winners and losers do not follow the same pattern as the momentum profits along the asset growth ranks. It appears that expected growth risk does not provide a satisfying explanation for the interaction between asset growth and momentum.

Next, we test whether the higher momentum profits in high asset growth groups can be explained by macroeconomic risk exposure. Cooper and Priestley (2011) document that the average return difference between low and high asset growth portfolios is, to a large extent, captured by the factor loadings that these portfolios have on the macroeconomic CRR factors. The result implies that firms that invest more also have lower systematic risk translating into lower average expected returns. While theory is silent about why systematic risk spreads between prior winners and prior losers should be larger among firms that have invested more, one can still hypothesize that more dynamic firms, characterized by high levels of investment, have better possibilities to choose from a wide variety of investment projects that have different exposures to systematic risk. This could lead to a wider spread in expected returns within the high asset growth groups, manifesting itself in stronger apparent return momentum than within the less extreme asset growth groups.

Our test setup follows Griffin, Xiuqing, and Spencer (2003) and Liu and Zhang (2008). We first estimate risk premiums for the five CRR factors. We then use these risk premiums together with the factor loadings on our 5-1 momentum portfolios to calculate the momentum profits that are predicted by these portfolios' macroeconomic risk exposures.

We find that while the macroeconomic risk model indeed predicts higher momentum returns in the high asset growth groups, it still fails to provide a full explanation of the interaction between asset growth and momentum. It also fails miserably in explaining the high momentum returns in asset growth group 1. Full details of these estimations and results are in the Internet

Appendix.

3.3 Relation to Behavioral Theories

We conclude with a brief overview of how our results reconcile with existing behavioral explanations for momentum. In short, we do not believe that current behavioral theories can fully account for our results. We refer the reader to our Internet Appendix for a thorough discussion of this theme.

A number of papers, among them Barberis et al. (1998) and Daniel et al. (1998), have argued that momentum profits arise because investors are plagued by systematic cognitive biases when they have to interpret the implications of new information. Furthermore, the behavioral biases that drive momentum should be more important for stocks with high information uncertainty than for stocks with low uncertainty. Thus, it could be that the interaction between asset growth and momentum is produced by the possibility that firms having experienced large changes to their asset bases are characterized by more value uncertainty and, consequently, by stronger momentum. We note, however, that we control in section 2.4 for market value, book-to-market, turnover and return volatility and find that the interaction between asset growth and momentum remains robust. All of these variables have been used as proxies for information uncertainty in earlier literature (Daniel and Titman, 1999; Jiang, Lee, and Zhang, 2005; Zhang, 2006). The robustness of the asset growth/momentum interaction in stocks otherwise classified as having low information uncertainty leads us to believe that information uncertainty is not the sole driving force behind our results.

Neither do our time-series tests of momentum profits described in section 2.5 lend direct support to the possibility that current behavioral-based theories would fully explain our results. If the slow market response to information is due to psychological biases, such as overconfidence,

then the mechanisms that result in momentum should largely be confined to periods of high investor overconfidence. We find that even though pessimistic investor sentiment appears to have a dampening effect on the magnitude of momentum profits, the interaction between asset growth and momentum remains robust when controlling for investor sentiment.

In our Internet Appendix, we also investigate the post-earnings-announcement drifts (PEAD) within the asset growth deciles. As noted by Hirshleifer (2001), the behavioral models mentioned above use exactly the same arguments for explaining momentum as they do for explaining the PEAD. Consequently, if magnified psychological biases were driving the high momentum returns in the extreme asset growth groups, we would also expect the returns on a PEAD strategy to be at their highest in those groups. However, our results, reported in the Appendix, do not provide strong evidence that returns on PEAD strategies are concentrated in the extreme asset growth deciles. This leads us to believe that cognitive biases among investors cannot fully explain our findings—rather, some other mechanisms seem to be at play.

4 Conclusions

In this paper, we establish a significant and robust connection between firm-level asset changes and stock price momentum. Asset growth retains its predictive ability when controlling for previously documented cross-sectional predictors of momentum and its predictive power often dominates that of other variables. Momentum profits are statistically significant and economically meaningful among firms that have experienced large asset expansions or contractions, but they are small and often insignificant among firms that have not experienced a drastic change in their asset bases. Our cross-sectional analysis shows that firm-level asset growth rates are not only predictors of future abnormal returns as in Cooper et al. (2008) but also strong predictors of short-term return momentum.

In addition to the cross-sectional evidence, we document a positive time-series relation between asset growth and price momentum, showing that average momentum returns increase almost monotonically with aggregate asset growth. The interaction between asset growth and momentum persists in various market states where prior literature has documented an absence of momentum profits. Furthermore, the results indicate that the time-series effect of aggregate asset growth subsumes the explanatory power of previously documented determinants of momentum related to the business cycle and aggregate investor sentiment. This finding provides strong complementary evidence to support our view that asset growth is an important determinant of the momentum effect.

The results have implications for the existing theories aiming to explain stock price momentum. We argue that most of the existing theoretical literature does not offer clear answers as to why firm investment, measured by asset growth, should be connected to return momentum. We subject our results to several robustness tests and find that the interaction between asset growth and momentum cannot be fully explained by pricing factors derived from investment-based asset pricing, by expected growth risk or by macroeconomic risk exposures. Furthermore, we argue, based on our results controlling for the effects of information uncertainty and investor overconfidence, that the behavioral theories of Barberis et al. (1998) and Daniel et al. (1998) cannot fully account for the interaction between asset growth and momentum. Our results on the post-earnings-announcement drifts within the asset growth deciles also suggest that the interaction between asset growth and stock price momentum may not originate solely from a systematic failure of information to flow correctly and completely into stock prices.

Our empirical results do, however, produce a pattern aligned with theoretical predictions of the model by Hackbarth and Johnson (2012). Even though the aim of our paper has not been to offer a formal test of their model, our independent empirical results nevertheless suggest that

a real options framework, such as the one considered by Hackbarth and Johnson (2012), can be fruitful for understanding asset return dynamics. This is important, because the cross-sectional return premiums uncovered in the literature appear to be linked to each other. Cooper et al. (2008) have linked asset growth to an investment premium, whereas our results show that asset growth is linked to the momentum premium. Furthermore, the results in Asness et al. (2012) indicate an underlying common structure for the momentum premium and the value premium in that they are negatively correlated. A promising avenue for future research would thus be to analyze asset growth in a real options framework, and see to what extent the momentum premium, the value premium and the investment premium can be linked to asset growth within a unified model.

References

- Ang, A., Hodrick, R. J., Xing, Y. and Zhang, X. (2006) The cross-section of volatility and expected returns. *Journal of Finance* **61**, 259–299.
- Antoniou, C., Doukas, J. and Subrahmanyam, A. (2010), Sentiment and momentum. Forthcoming, *Journal of Financial and Quantitative Analysis*.
- Arena, M., Haggard, K. S. and Yan, X. (2008) Price momentum and idiosyncratic volatility. *The Financial Review* **43**, 159–190.
- Asness, C., Moskowitz, T. and Pedersen, L. H. (2012), Value and momentum everywhere. Forthcoming, *Journal of Finance*.
- Avramov, D., Chordia, T., Jostova, G. and Philipov, A. (2007) Momentum and credit rating. *Journal of Finance* **62**, 2503–2520.
- Baker, M. and Wurgler, J. (2006) Investor sentiment and the cross-section of stock returns. *Journal of Finance* **61**, 1645–1680.
- Baker, M. and Wurgler, J. (2007) Investor sentiment in the stock market. *Journal of Economic Perspectives* **21**, 129–151.
- Bandarchuk, P. and Hilscher, J. (2013) Sources of momentum profits: Evidence on the irrelevance of characteristics. *Review of Finance* **17**.
- Barberis, N., Shleifer, A. and Vishny, R. (1998) A model of investor sentiment. *Journal of Financial Economics* **49**, 307–343.
- Berk, J., Green, R. C. and Naik, V. (1999) Optimal investment, growth options, and security returns. *Journal of Finance* **54**, 1553–1607.
- Carlson, M., Fisher, A. and Giammarino, R. (2004) Corporate investment and asset price dynamics: Implications for the cross-section of returns. *Journal of Finance* **59**, 2577–2603.
- Chan, L., Karceski, J., Lakonishok, J. and Sougiannis, T. (2008), Balance sheet growth and the predictability of stock returns. Working paper, University of Illinois at Urbana-Champaign.
- Chen, L., Novy-Marx, R. and Zhang, L. (2010), An alternative three-factor model. Unpublished Working Paper, Washington University in St. Louis.
- Chen, N.-F., Roll, R. and Ross, S. A. (1986) Economic forces and the stock market. *Journal of Business* **59**, 383–403.
- Chordia, T. and Shivakumar, L. (2002) Momentum, business cycle, and time-varying expected returns. *Journal of Finance* **57**, 985–1019.
- Cooper, I. (2006) Asset pricing implications of nonconvex adjustment costs and irreversibility of investment. *Journal of Finance* **61**, 139–170.
- Cooper, I. and Priestley, R. (2011) Real investment and risk dynamics. *Journal of Financial Economics* **101**, 182–205.

- Cooper, M., Gulen, H. and Schill, M. (2008) Asset growth and the cross-section of stock returns. *Journal of Finance* **63**, 1609–1651.
- Cooper, M., Gutierrez, C. and Hameed, A. (2004) Market states and momentum. *Journal of Finance* **49**, 1345–1365.
- Daniel, K. D., Grinblatt, M., Titman, S. and Wermers, R. (1997) Measuring mutual fund performance with characteristic-based benchmarks. *Journal of Finance* **52**, 1035–1058.
- Daniel, K. D., Hirshleifer, D. and Subrahmanyam, A. (1998) Investor psychology and security market under- and overreactions. *Journal of Finance* **53**, 1839–1886.
- Daniel, K. D. and Titman, S. (1999) Market efficiency in an irrational world. *Financial Analysts' Journal* **55**, 28–40.
- Datar, V., Naik, N. and Radcliffe, R. (1998) Liquidity and stock returns: An alternative test. *Journal of Financial Markets* **1**, 203–219.
- Davis, J., Fama, E. F. and French, K. R. (2000) Characteristics, covariances, and average returns: 1929–1997. *Journal of Finance* **55**, 389–406.
- Fama, E. and French, K. (1993) Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* **33**, 3–56.
- Fama, E. and French, K. (1996) Multifactor explanations of asset pricing anomalies. *Journal of Finance* **51**, 55–84.
- Fama, E. and French, K. (2006) Profitability, investment and average returns. *Journal of Financial Economics* **82**, 491–518.
- Fama, E. and French, K. (2008) Dissecting anomalies. *Journal of Finance* **63**, 1653–1678.
- Fama, E. and MacBeth, J. (1973) Risk, return and equilibrium: Empirical tests. *Journal of Political Economy* **81**, 607–636.
- Garlappi, L. and Yan, H. (2011) Financial distress and the cross-section of equity returns. *Journal of Finance* **66**, 789–822.
- Gomes, J., Kogan, L. and Zhang, L. (2003) Equilibrium cross-section of returns. *Journal of Political Economy* **111**, 693–732.
- Griffin, J., Xiuqing, J. and Spencer, M. (2003) Momentum investing and business cycle risk: Evidence from pole to pole. *Journal of Finance* **58**, 2515–2547.
- Hackbarth, D. and Johnson, T. (2012), Real options and risk dynamics. Unpublished working paper, University of Illinois, Urbana-Champaign.
- Hirshleifer, D. (2001) Investor psychology and asset pricing. *Journal of Finance* **64**, 1533–1597.
- Hong, H., Lim, T. and Stein, J. (2000) Bad news travels slowly: Size, analyst coverage and the profitability of momentum strategies. *Journal of Finance* **55**, 265–295.
- Hong, H. and Stein, J. C. (1999) A unified theory of underreaction, momentum trading and overreaction in asset markets. *Journal of Finance* **54**, 2143–2184.

- Jegadeesh, N. (1990) Evidence of predictable behavior of security returns. *Journal of Finance* **45**, 881–898.
- Jegadeesh, N. and Titman, S. (1993) Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance* **48**, 65–91.
- Jegadeesh, N. and Titman, S. (2001) Profitability of momentum strategies: An evaluation of alternative explanations. *Journal of Finance* **56**, 699–720.
- Jiang, G., Lee, C. and Zhang, Y. (2005) Information uncertainty and expected returns. *Review of Accounting Studies* **10**, 185–221.
- Johnson, T. (2002) Rational momentum effects. *Journal of Finance* **57**, 585–608.
- Lee, C. and Swaminathan, B. (2000) Price momentum and trading volume. *Journal of Finance* **55**, 2017–2069.
- Liu, L. and Zhang, L. (2008) Momentum profits, factor pricing and macroeconomic risk. *Review of Financial Studies* **21**, 2417–2448.
- Newey, W. K. and West, K. D. (1987) A simple positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* **55**, 703–708.
- Sagi, J. and Seasholes, M. (2007) Firm-specific attributes and the cross-section of momentum. *Journal of Financial Economics* **84**, 389–434.
- Zhang, F. (2006) Information uncertainty and stock returns. *Journal of Finance* **61**, 105–136.
- Zhang, L. (2005) The value premium. *Journal of Finance* **60**, 67–103.

Table I

Portfolios sorted on past asset growth and past returns

Panel A shows descriptive statistics for ten portfolios sorted on the yearly percentage change in balance sheet total assets (Compustat data item 6), denoted AG. We exclude stocks with a market value that at the end of June each year falls below the 20th percentile of market capitalization of NYSE stocks. The sample period for the holding period returns is from July 1968 to June 2006. The reported equal- and value-weighted portfolio returns are average monthly holding period returns (HPRet (EW), HPRet (VW)). The significance of the mean raw return difference between the extreme low and high asset growth portfolios (1-10) is estimated with a Newey-West (1987) t-statistic. The Fama-French (1993) three-factor alphas (FF3) are estimated to evaluate the return difference between the portfolios 1 and 10. The standard deviation of equal-weighted returns (Std (EW)) and average market shares (Market share) are reported for each decile. Descriptive statistics are reported for the time-series averages of value-weighted past returns (PastRet, measured from months 2 to 12 prior to portfolio formation), book-to-market (BM) and lagged asset growth rates (AG). Panel B documents results for five portfolios sorted on past 11 month returns using NYSE prior (2-12) return breakpoints, with a one month lag between the formation and holding periods. The holding period is one month. Panel C reports a time-series average of the cross-sectional correlations between the relevant firm-specific variables asset growth (AG), book-to-market (BM), shares traded divided by shares outstanding (Turnover), idiosyncratic volatility calculated relative to the Fama-French (1993) three-factor model (IVol) and market capitalization (Size).

Panel A. Portfolios sorted on asset growth—20% screen for size

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	1-10	t	FF3	t
HPRet (EW)	1.25	1.30	1.32	1.24	1.25	1.22	1.22	1.11	1.00	0.59	0.66	[4.24]	0.45	[3.06]
HPRet (VW)	1.17	1.20	1.07	1.02	1.00	0.99	1.07	0.87	0.88	0.64	0.53	[3.02]	0.30	[1.99]
Std (EW)	6.1	5.0	4.7	4.5	4.7	4.8	5.1	5.4	6.2	7.2				
Market share	5.6	8.5	10.8	11.5	12.4	12.7	12.3	10.0	9.4	6.9				
AG (VW)	-9.7	0.1	3.4	6.1	8.7	11.6	15.2	20.5	30.7	88.7				
BM (VW)	0.76	0.81	0.74	0.66	0.61	0.55	0.52	0.47	0.43	0.42				
PastRet (VW)	10.3	11.1	11.8	12.3	11.6	11.9	11.0	11.8	12.4	12.6				

Panel B. Portfolios sorted on past returns—20% screen for size

	P1	P2	P3	P4	P5	P5-P1	t	FF3	t
HPRet (EW)	0.66	1.01	1.11	1.27	1.56	0.90	[4.17]	1.14	[4.87]
HPRet (VW)	0.62	0.89	0.82	1.02	1.33	0.71	[2.90]	0.90	[3.45]
Std (EW)	7.3	5.3	4.8	4.7	5.8				
Market share (%)	10.9	20.3	23.0	24.1	21.7				
Asset growth (VW)	25.2	16.2	14.2	13.9	17.9				
Past return (2-12) (VW)	-25.5	-4.6	9.3	24.6	56.7				

Panel C. Correlations between firm-specific variables

Series	Asset growth	BM	Turnover	Ivol	Size
Asset growth	1	-0.15	0.13	0.15	-0.01
BM		1	-0.06	-0.09	-0.09
Turnover			1	0.50	-0.08
Ivol				1	-0.21
Size					1

Table II

Momentum within asset growth groups

Panel A presents average returns on portfolios sorted independently on balance sheet asset growth and past return. At the end of June each year, stocks are sorted into ten groups based on their asset growth rates in the previous year. Within each asset growth decile, stocks are ranked at the end of each sample month based on their past 11-months returns, excluding the month preceding the first holding period month. The stocks are then grouped into five portfolios based on the NYSE prior return breakpoints. The momentum strategy (P5-P1) involves buying the past winner portfolio (P5) and selling the past loser portfolio (P1). The holding period is one month. The table shows the average monthly portfolio returns during the holding period as well as the average momentum strategy returns on the 5-1 hedge portfolio. The corresponding momentum profits adjusted with the Fama-French (1993) three-factor model are also reported. Newey and West (1987) t-values are reported in brackets. Only stocks with a market value that at the end of June exceeds the 20th NYSE size percentile are included in the analysis. Panel B.I presents the 5-1 raw momentum returns as in Panel A, but without any screens for firm size. Average raw momentum returns using longer holding periods of three and six months are reported in Panel B.II. The sample is from July 1968 to June 2006. Panel C studies the impact of excluding firms with a Standard and Poor's Long-Term Domestic Issuer credit rating lower than B from the sample. The sample is from July 1986 to June 2006.

Panel A.I Momentum within asset growth groups—equal-weighted returns

Asset growth	Past return groups					P5-P1	t	FF3	t
	P1	P2	P3	P4	P5				
Low	0.62	1.20	1.25	1.45	1.65	1.03	[4.51]	1.36	[5.65]
2	1.19	1.15	1.31	1.35	1.44	0.26	[0.96]	0.43	[1.59]
3	1.25	1.29	1.16	1.37	1.60	0.34	[1.30]	0.52	[1.92]
4	1.01	1.12	1.21	1.18	1.54	0.52	[2.18]	0.80	[2.92]
5	1.05	1.14	1.16	1.30	1.59	0.53	[2.09]	0.86	[3.33]
6	1.13	1.03	1.17	1.28	1.48	0.35	[1.42]	0.57	[2.21]
7	0.86	1.00	1.08	1.37	1.67	0.81	[3.59]	1.00	[3.95]
8	0.54	0.94	1.04	1.24	1.60	1.06	[4.35]	1.36	[5.36]
9	0.34	0.64	0.91	1.22	1.68	1.34	[5.06]	1.64	[6.10]
High	-0.13	0.44	0.50	0.86	1.39	1.52	[5.68]	1.83	[6.39]

Panel A.II Momentum within asset growth groups—value-weighted returns

Asset growth	Past return groups					P5-P1	t	FF3	t
	P1	P2	P3	P4	P5				
Low	0.60	1.15	1.09	1.32	1.39	0.78	[2.94]	1.19	[4.22]
2	1.22	1.01	1.09	1.36	1.24	0.02	[0.05]	0.12	[0.36]
3	1.29	1.37	0.87	1.15	1.44	0.15	[0.42]	0.23	[0.60]
4	1.03	0.96	0.98	0.99	1.28	0.25	[0.74]	0.67	[1.95]
5	1.18	0.95	0.84	1.01	1.45	0.27	[0.84]	0.54	[1.70]
6	0.73	0.84	1.07	1.03	1.22	0.50	[1.77]	0.64	[2.07]
7	0.62	1.05	0.82	1.14	1.44	0.81	[2.92]	1.00	[3.43]
8	0.25	0.77	0.72	0.95	1.40	1.15	[3.82]	1.43	[4.59]
9	0.28	0.41	0.65	1.00	1.54	1.27	[4.28]	1.50	[4.45]
High	-0.16	0.35	0.61	0.62	1.18	1.34	[4.33]	1.71	[5.28]

Panel B. Robustness tests

I. No screen for size						II. 20% screen for size			
Asset growth	11-1-1					11-1-3		11-1-6	
	P5-P1	t	P5-P1	t	AG	P5-P1	t	P5-P1	t
Low	-0.19	[-0.76]	0.65	[2.12]	-23.0	0.82	[3.35]	0.57	[2.38]
2	0.27	[1.16]	0.92	[3.19]	-6.2	0.31	[1.21]	0.27	[1.13]
3	0.09	[0.44]	-0.02	[-0.06]	-0.5	0.36	[1.50]	0.33	[1.47]
4	0.57	[3.06]	0.42	[1.30]	3.1	0.68	[2.88]	0.74	[3.15]
5	0.85	[4.10]	0.26	[0.83]	6.2	0.42	[1.72]	0.38	[1.61]
6	0.69	[3.32]	0.39	[1.28]	9.4	0.40	[1.66]	0.33	[1.46]
7	0.73	[3.59]	0.79	[3.12]	13.3	0.76	[3.36]	0.56	[2.49]
8	1.01	[4.80]	1.00	[3.33]	18.9	0.86	[3.66]	0.78	[3.58]
9	1.39	[6.86]	1.49	[5.44]	29.8	1.22	[4.86]	0.99	[4.17]
High	1.39	[5.75]	1.24	[4.26]	96.9	1.30	[5.04]	1.12	[4.54]
Weighting	EW		VW			EW		EW	

Panel C. Impact of credit rating, July 1986 June 2006

20% screen for size									
Asset growth	11-1-1								
	P5-P1	t	P5-P1	t		P5-P1	t	P5-P1	t
Low	1.00	[3.04]	1.02	[2.92]		0.62	[1.82]	0.67	[1.79]
2	0.00	[0.00]	0.07	[0.20]		0.20	[0.46]	0.29	[0.65]
3	0.30	[0.85]	0.28	[0.79]		0.24	[0.62]	0.20	[0.54]
4	0.42	[1.18]	0.38	[1.07]		0.24	[0.54]	0.18	[0.42]
5	0.61	[1.58]	0.64	[1.68]		0.63	[1.25]	0.74	[1.53]
6	-0.11	[-0.32]	-0.08	[-0.25]		0.32	[0.69]	0.34	[0.78]
7	0.57	[1.80]	0.61	[2.02]		1.17	[2.41]	1.33	[2.85]
8	0.68	[2.11]	0.62	[1.92]		1.16	[2.52]	1.05	[2.33]
9	1.16	[2.91]	1.10	[2.86]		1.46	[2.94]	1.30	[2.78]
High	1.25	[3.40]	1.21	[3.33]		1.86	[3.55]	1.80	[3.43]
Sample	Rated and unrated		Rated and unrated excluding rated below B			Only rated		Only rated excluding rated below B	

Table III

Controlling for other drivers of momentum

Panel A presents average monthly momentum returns from July 1968 (July 1986 when controlling for credit ratings) to June 2006 within asset growth groups while controlling for other firm-specific variables. First, stocks are sorted into three groups based on the magnitude of the control variable. Then in an independent sort, stocks are arranged into three groups based on their balance sheet asset growth rates. Finally, within each of the nine asset growth/control variable groups, stocks are ranked on their past 11-month returns (excluding the month preceding the first holding period month) into five portfolios based on the NYSE prior return breakpoints. In Panel A.I. the size breakpoints are defined as the 20th and 50th percentiles of market capitalization for NYSE stocks. In Panel A.II book value is calculated as in Davis, Fama, and French (2000) and divided by the market value at the end of June in year t to obtain book-to-market ratios. Negative book values are excluded. In Panel A.III., trading volume is the average daily turnover (number of shares traded / number of shares outstanding) within the 11-month portfolio formation period. Nasdaq stocks are excluded. Idiosyncratic volatility (Panel A.IV.) is relative to the Fama-French (1993) three-factor model using daily returns within the 11-month formation period. In Panel A.V. Standard and Poor's Long-Term Domestic Issuer Credit Ratings are used as a control variable. In contrast to the previous sorts, past 11-month return breakpoints are calculated within each credit rating/asset growth group. Panel B reports firm-level Fama-MacBeth (1973) regressions of monthly stock returns on lagged empirical determinants of cross-sectional returns and on terms that capture how the different control variables interact with momentum. In the regression results, we apply the 20% screen for size to the stocks in our sample. In specifications that include turnover, Nasdaq stocks are excluded, and the sample starts from start from July 1986 when credit rating is included. Newey-West (1987) t -statistics are given in brackets.

Panel A: Three-way independent portfolio sorts – Equal-weighted returns

	I. Size			II. Book-to-market			III. Turnover			IV. Idiosyncratic volatility					
	Micro caps		Big stocks	High BM		Low BM	Low TO		Med TO	High TO	Low IVOL		Med IVOL		High IVOL
Low AG	0.04	1.04	0.09	0.21	0.15	0.36	0.04	0.76	0.88	0.46	1.02	-0.78			
	[0.19]	[4.57]	[0.30]	[0.90]	[0.72]	[1.62]	[0.13]	[3.04]	[3.46]	[3.14]	[4.99]	[-2.21]			
Med AG	0.96	0.74	0.26	0.58	0.78	0.92	0.86	0.59	1.14	0.38	0.99	0.99			
	[5.44]	[3.49]	[1.10]	[3.08]	[4.35]	[4.50]	[4.32]	[2.65]	[4.99]	[3.23]	[5.64]	[3.39]			
High AG	1.26	1.32	1.12	1.20	1.12	1.59	0.94	1.18	1.51	0.86	1.51	1.15			
	[6.35]	[5.57]	[4.40]	[4.78]	[5.75]	[8.02]	[4.34]	[5.41]	[6.21]	[5.36]	[7.38]	[3.43]			
Spread in momentum (EW)	1.22	0.28	1.03	0.99	0.97	1.23	0.91	0.42	0.63	0.40	0.49	1.94			
	[6.56]	[1.53]	[4.49]	[3.99]	[4.74]	[6.68]	[3.49]	[1.95]	[3.06]	[2.64]	[2.88]	[5.64]			

V. Credit rating

	High CR		Med CR		Low CR	
Low AG	-0.13		-0.04		1.13	
	[-0.47]		[-0.13]		[1.92]	
Med AG	0.12		0.78		1.59	
	[0.46]		[2.12]		[2.58]	
High AG	0.42		0.89		2.95	
	[1.36]		[2.24]		[4.69]	
Spread in momentum (EW)	0.55		0.93		1.82	
	[2.55]		[3.05]		[2.67]	

Panel B: Firm-level Fama-MacBeth (1973) regressions

	1	2	3	4	5	6	7
Intercept	2.559 [4.04]	2.446 [3.98]	2.558 [4.06]	2.554 [4.07]	2.395 [3.91]	1.558 [1.43]	1.577 [1.76]
AG	-0.003 [-3.60]	-0.002 [-2.64]	-0.003 [-3.60]	-0.003 [-3.40]	-0.003 [-3.11]	-0.003 [-1.68]	-0.003 [-2.11]
MV	-0.080 [-2.15]	-0.075 [-1.98]	-0.080 [-2.15]	-0.080 [-2.17]	-0.074 [-1.95]	-0.031 [-0.46]	-0.038 [-0.64]
BM	0.178 [2.31]	0.165 [2.34]	0.181 [2.37]	0.194 [2.27]	0.195 [2.38]	0.082 [0.82]	0.062 [0.58]
ret11	0.009 [5.28]	0.009 [5.35]	0.008 [4.83]	0.008 [3.94]	0.010 [4.17]	0.009 [2.87]	0.010 [2.35]
IVOL	-0.218 [-2.01]	-0.163 [-1.70]	-0.211 [-1.95]	-0.208 [-1.95]	-0.153 [-1.59]	-0.044 [-0.31]	-0.043 [-0.33]
TO		-0.660 [-3.05]			-0.636 [-2.32]		0.090 [0.34]
CR						-0.265 [-1.37]	-0.300 [-1.45]
ret11 × AG			0.005 [2.86]	0.04 [2.22]	0.006 [2.89]	0.004 [2.12]	0.008 [2.72]
ret11 × BM				-0.132 [-1.43]	-0.122 [-1.00]		0.211 [0.95]
ret11 × TO					-0.553 [-1.56]		0.061 [0.16]
ret11 × CR						0.764 [1.50]	0.661 [1.44]

Table IV

Momentum profits conditioning on market states

Panel A groups the holding period months in expansionary and contractionary months based on the classifications made by the NBER. The table reports, for these two states of the economy, the average profits on 5-1 momentum strategies that use all stocks in the sample (applying a 20% screen for market capitalization) and on the momentum strategies within ten asset growth groups. To assess the market state in Panel B, we use the lagged 12-month cumulative returns on the CRSP value-weighted market index. If the 12-month lagged return on the index is positive (skipping one month before the holding period), a holding-period month is classified as an UP month, otherwise it is classified as a DOWN month. Panel B also reports the results for the DOWN market momentum profits when January 2001 is excluded from the sample (labeled DOWN*). In Panel C, the monthly sentiment index constructed by Baker and Wurgler (2006, 2007) is used to classify the sample months into pessimistic and optimistic periods. A moving average of the sentiment values is calculated for the three months prior to the holding period. If this average sentiment level falls within the bottom 33% of all months, the holding period month is classified as pessimistic, otherwise it is optimistic. The sample is from July 1968 to June 2006. Newey-West (1987) t-statistics are in brackets.

	A.			B.				C.						
	Expand	t	Contract	t	UP	t	DOWN	t	DOWN*	t*	Optimistic	t	Pessimistic	t
All stocks	0.96	[4.20]	0.54	[0.85]	1.19	5.80	-0.02	[-0.02]	0.37	[0.64]	1.13	[4.04]	0.45	[1.02]
Low AG	1.08	[4.29]	0.75	[1.15]	1.35	[5.60]	0.02	[0.04]	0.28	[0.45]	1.22	[4.27]	0.70	[1.40]
2	0.29	[1.04]	0.04	[0.05]	0.65	[2.68]	-0.99	[-1.36]	-0.54	[-0.80]	0.43	[1.28]	-0.15	[-0.29]
3	0.53	[2.29]	-0.80	[-0.77]	0.88	[3.69]	-1.35	[-1.72]	-1.05	[-1.34]	0.43	[1.31]	0.16	[0.31]
4	0.61	[2.37]	-0.01	[-0.01]	0.92	[3.80]	-0.73	[-1.08]	-0.38	[-0.61]	0.89	[3.22]	-0.09	[-0.18]
5	0.60	[2.13]	0.13	[0.17]	0.83	[3.10]	-0.40	[-0.69]	-0.07	[-0.13]	0.83	[2.49]	-0.03	[-0.06]
6	0.44	[1.66]	-0.21	[-0.33]	0.69	[2.81]	-0.75	[-1.22]	-0.51	[-0.87]	0.61	[1.96]	-0.18	[-0.34]
7	0.99	[3.94]	-0.23	[-0.38]	1.15	[4.79]	-0.26	[-0.42]	0.08	[0.15]	0.90	[3.03]	0.67	[1.62]
8	1.16	[4.48]	0.43	[0.73]	1.35	[5.11]	0.13	[0.23]	0.46	[0.96]	1.09	[3.35]	0.93	[2.18]
9	1.39	[4.93]	1.05	[1.37]	1.77	[7.00]	0.00	[0.01]	0.42	[0.68]	1.64	[4.93]	0.69	[1.46]
High AG	1.54	[5.55]	1.40	[1.52]	1.69	[5.99]	0.97	[1.29]	1.52	[2.23]	1.78	[5.07]	1.03	[2.24]
N	391		65		346		110		109		300			150

Table V

Momentum profits conditioned on aggregate average asset growth

Table V examines momentum profits using aggregate firm expansion as a conditioning variable. For each firm, the annual asset growth rate is calculated from the balance sheet total assets during the fiscal quarter ending in calendar quarter t relative to the total assets in quarter $t-4$. Then, an average is taken for each quarter t over all firms for which the asset growth rates are available. We obtain quarterly returns on 10 momentum portfolios. The quarterly momentum return is the return difference between the winner and loser portfolios. The asset growth rates at time t are matched with the 10-1 momentum returns during quarter $t + 1$. Then, the time series of aggregate asset growth rates are sorted into four groups based on their magnitudes allowing us to study the resulting momentum profits within each of these four market states. Panel A presents the mean quarterly momentum returns and their corresponding t -statistics when the returns are conditioned on aggregate asset growth. The last column shows the quarterly mean returns transformed to a monthly basis. In Panel B, we regress the quarterly 10-1 momentum returns on one-quarter lagged aggregate asset growth rates and other variables that have previously been shown to predict momentum payoffs. The variables are a recessionary dummy as defined by the NBER, lagged market sentiment, lagged 12-month market return and the square of lagged market returns. Model 5 restricts the sample to those observations when the lagged market return has been positive.

Panel A. Quarterly momentum profits across periods
of low and high aggregate asset growth rates

Aggregate asset growth	P10-P1	t	P10-P1 ^{month}
Low	0.55	[0.27]	0.18
2	2.13	[1.05]	0.71
3	4.50	[4.28]	1.48
High	5.30	[4.28]	1.74
High-Low	4.76	[1.97]	

Panel B: Time-series regressions					
Model	1	2	3	4	5
Intercept	-1.85	-0.87	-1.22	1.24	3.55
	[-0.73]	[-0.33]	[-0.51]	[0.46]	[1.54]
$AG_{t-1}^{quarter}$	0.30	0.26	0.26	0.14	0.22
	[2.45]	[2.11]	[2.20]	[1.12]	[2.82]
recdummy		-2.63			
		[-0.73]			
sentiment $_{t-1}$			1.01		
			[1.46]		
market $_{t-1}$				0.28	-0.16
				[3.45]	[-1.08]
market $_{t-1}^2$				-0.01	0.00
				[3.51]	[0.21]

Table VI

Momentum profits when controlling for profitability

Panel A examines momentum profits across asset growth deciles when controlling for the Chen, Novy-Marx and Zhang (2010) factors. For each asset growth group, we regress the time series of the equal-weighted 5-1 momentum returns on a constant and the CNZ-factors. The sample starts from January 1972. Panel A reports the raw returns on the momentum portfolios, the CNZ-alphas and the corresponding t-statistics. In Panel B, we rank firms at the end of June each year in three groups based on their past asset growth rates. We apply a 20% screen for size (based on NYSE-breakpoints) and exclude firms falling within the lowest decile of asset growth rates. Then, at the beginning of each month, we independently rank firms in three groups based on their quarterly ROAs. ROA is defined as the firm's most recently reported quarterly income before extraordinary items (Compustat item IBQ) divided by the total assets (ATQ) in the previous quarter. We form nine asset growth/ROA groups. Finally, within each group, we rank stocks at the end of each sample month based on their past 11-month returns, and then group these stocks into 5 portfolios based on the NYSE prior return breakpoints. The panel reports the average winner-minus-loser returns and their corresponding t-statistics. We start the sample from July 1980.

Controlling for Chen, Novy-Marx and Zhang factors and for return on assets (ROA)								
A. Raw returns and CNZ-alphas on 5-1 momentum portfolios					B. Three-way independent portfolio sorts			
Asset growth	P5-P1	t	CNZ	t		Low ROA	Med ROA	High ROA
Low	1.05	[4.30]	0.64	[1.88]	Low AG	0.62	0.00	0.29
2	0.34	[1.21]	-0.32	[-0.94]		[1.72]	[0.00]	[1.05]
3	0.39	[1.58]	0.04	[0.14]	Med AG	0.39	0.40	-0.08
4	0.53	[2.06]	0.27	[0.63]		[1.07]	[1.35]	[-0.23]
5	0.57	[2.08]	0.21	[0.53]	High AG	0.97	0.84	1.13
6	0.30	[1.18]	0.03	[0.09]		[2.69]	[2.52]	[3.82]
7	0.81	[3.37]	0.47	[1.34]				
8	1.09	[4.35]	0.75	[2.00]	Spread in raw	0.35	0.84	0.84
9	1.36	[4.85]	1.02	[2.65]	momentum returns	[1.23]	[2.81]	[2.84]
High	1.53	[5.36]	0.93	[2.27]				
					Spread in	0.01	0.89	1.07
					CNZ alphas	[0.04]	[2.87]	[3.25]